Softwarica COLLEGE OF IT & E-COMMERCE | in collaboration with Coventry University

Student Assignment Brief

CONFIDENTIAL DOCUMENT

This document is intended solely for Softwarica College of IT & E-Commerce students for their own use in completing their assessed work for this module. It must not be passed to third parties or posted on any website.

Contents

- Assignment Information
- Assessed Module Learning Outcomes
- Assignment Task
- Marking and Feedback
- Assignment Support and Academic Integrity
- Assessment Marking Criteria

Assignment Information

Module Name:	Machine Learning
Module Code:	ST7072CEM
Assignment Title:	STW7072CEM Machine Learning March Intake 2026 Coursework CW (Regular)
Assignment Due:	28-September-2026, 05:00PM

Assignment Credit:	20
Word Count:	6000
Assignment Type:	Coursework : Individual Report
Grading:	Percentage Grade (Applied Core Assessment).

Assessment Overview

You will be provided with an overall grade between 0% and 100%. You have one opportunity to pass the assignment at or above 40%.

Important Notice

The work you submit for this assignment must be your **own independent work**. More information is available in the 'Assignment Task' section of this assignment brief.

Assessed Module Learning Outcomes

The Learning Outcomes for this module align with the marking criteria which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task. The following module learning outcomes are assessed in this task

1. Examine the fundamental concepts of machine learning, their implementation and application.
2. Prepare appropriate preparation of data sets and evaluate the performance of different learning algorithms on these data sets.
3. Appraise the different learning methodologies, their associated algorithms and their appropriateness to real-world problems.
4. Select and apply learning algorithms to various practical scenarios and evaluate their performance.
5. Critique trends in the machine learning developments.

Assignment Task

Task:
This assignment will involve the development of practical work, software planning for an intelligent application prototype, and the development of that prototype. You have been issued with exercises associated with lab work throughout the module.
The planning and development of the intelligent application prototype will make use of the knowledge and materials that you have been collecting. Intelligent application

prototype in the python programming language using Jupyter Notebook or Google Colab.

Project Prototype: Using Machine Learning Methods to Address Clustering and Classification Problems in Real-World Contexts.
Context: You were familiar with many machine learning methods, related ideas, and applications during this module. Several classification techniques were studied, including Generalized Logistic Regression, Linear Discriminant Analysis, Optimized K-nearest Neighbor, Bayesian and Statistical Methods, Support Vector Machines, and Decision Trees. In addition, we examined feature selection and extraction techniques like Principal Component Analysis as well as clustering algorithms like K-means. You must choose an application that relates to a classification, clustering, or anomaly detection problem for this project and investigate the most effective way to use machine learning methods to solve it. You are open to choosing any datasets available.
Sources: https://www.kaggle.com or https://archive.ics.uci.edu/ and need to apply 3 different classification or clustering techniques on it.
The purpose of this coursework is to
● Analyze the core ideas of machine learning, as well as how they are used.
● Ensure that a dataset is properly prepared, and then assess how well various learning algorithms work on it.
● Learn how to choose machine learning methods to address a real-world classification or clustering issue.
The following sections should generally be included in the paper:

1. Abstract
2. The problem is introduced along with a brief literature review of related work in the introduction; if the literature review is longer, it is advised to be its own section, as this would be preferable.
3. Problem and Data set(s) description (where you describe the problem in detail. you want to solve and its significance)
4. Methods (where you shortly describe the machine learning methods and/or other methods employed to solve the problem) Experimental setup (including data pre-processing, feature selection and extraction, classification/clustering parameters)
5. Results
6. Discussion and Conclusions
7. References

These are general section headings that you can modify to fit the application or problem under investigation. Sections discussing fresh and unique developments or modifications of algorithms relevant to your work may be included.
Your code needs to be developed in the python programming language using Jupyter Notebook or Google Colab and submitted as a zip file. You should include a comprehensive set of instructions for the successful execution of your code as well as a

short screencast/video of your code running (or a link to the screencast/video if it is too large to include).
More information of how to write a paper is available at the following link: “Crafting Papers on Machine Learning”, by Pat Langley (which can be found here if the previous link does not work
http://www.machinelearning.ru/wiki/images/0/07/Langley00crafting.pdf).
You will need to follow the formatting guidelines of the IEEE Manuscript Template for Conference Proceedings (https://www.ieee.org/conferences/publishing/templates.html).
As mentioned above, you must choose a real-world classification/clustering issue and one or more suitable dataset(s).
Additionally, you can use the links below: UCI Machine Learning Repository: http://archive.ics.uci.edu/ml/; ICML 2019 accepted papers: https://icml.cc/Conferences/2019/Schedule?type=Poster; Kaggle competitions: http://www.kaggle.com/competitions; Stanford machine learning projects: http://cs229.stanford.edu/projects2013.html, http://cs229.stanford.edu/projects2012.html, http://cs229.stanford.edu/projects2011.html, http://cs229.stanford.edu/projects2016.html.

1. You will draft a proposal (no longer than one A4 page) outlining the project's title and must present your tutor with this plan in order to receive constructive criticism.
2. You will need to investigate and read related work in the next 2 days. You are welcome to submit an individually written short literature review of your findings in order to get formative feedback.
3. You must choose, implement, and apply the best machine learning algorithms to the chosen problem throughout the course of the next two to three weeks. If necessary, you must also execute data preprocessing.
4. You are welcome to report your progress to receive regular feedback from the module leader or tutor in the following 2 support weeks, via some scheduled meetings with the module leader.
5. You have to write up your final paper and submit it by the deadline specified on the first page.

Submission Instructions

Requirement	Details
File Naming	NAME_studentID
File Format	.docx/.pdf format

Submission Method	Campus 4.0 platform (submission link provided 2 weeks before deadline)

Marking and Feedback

How will my assignment be marked?

Your assignment will be marked by the Module Team using standardized criteria.

How will I receive grades and feedback?

Provisional marks will be released once internally moderated. Feedback will be provided alongside grades release within 2 weeks (10 working days).

What will I be marked against?

Details of the marking criteria for this task can be found in the Assessment Marking Criteria section at the end of this brief.

Grade Requirements

You must achieve 40% or above to pass this assessment. Ensure you understand the marking criteria for successful completion.

Assignment Support and Academic Integrity

Getting Help

If you have any questions about this assignment, please meet your respective module leader or teacher for more information.

Language Standards

You are expected to use effective, accurate, and appropriate language within this assessment task.

Academic Integrity

The work you submit must be your own. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work, **excluding Artificial Intelligence (AI)**. We use detection software and make routine checks for evidence of academic misconduct. Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found in Student handbook in Campus 4.0. All cases of suspected academic misconduct are referred to for investigation, the outcomes of which can have profound consequences to your studies.

Support for Students with Disabilities

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms, contact the Student Support Office for assistance.

Unable to Submit on Time?

If events prevent you from submitting on time, guidance on extenuating circumstances is available in the Student Handbook or from the Student Support Office.

Administration of Assessment

Module Leader Name:	Shrawan Thakur
Module Leader Email:	stw0049@softwarica.edu.np
Assignment Category:	Proposal
Attempt Type:	regular
Component Code:	cw

Assessment Marking Criteria

Criterion	Marks
Technical quality	
1. Rigor and extent of the experiments. 2. Correct application of the selected algorithms and suitability of the methods. 3. Data preparation - technical quality. 4. Extent of evidence of running the experiments provided in appendices.	10% 10% 10% 10%
Evaluation	
5. Discussion and evaluation of the findings. Why are the outcomes significant? What applications would the findings have for other academics or professionals? 6. Is this a "serious" issue or just a minor "toy" issue? In what ways does the paper improve the field of study?	15% 5%
7. **Social, ethical, legal, and professional considerations** related to the problem in question.	5%

This document is confidential and intended solely for enrolled students.